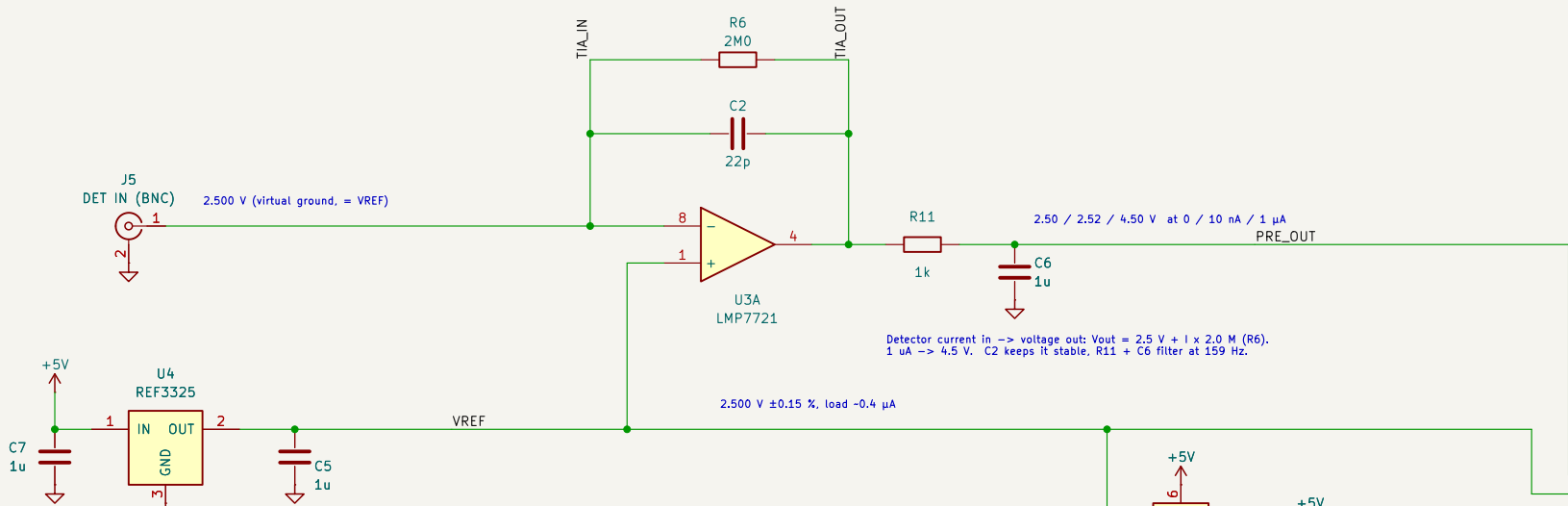


SIGNAL FLOW: detector -> preamp -> ADC -> Arduino -> USB -> laptop. Arduino -> DAC -> quadrupole supply. Trigger -> Arduino.

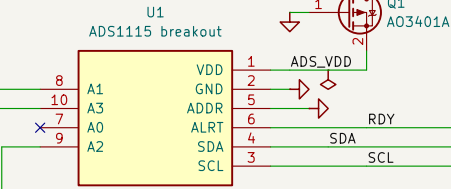
1 DETECTOR PREAMP



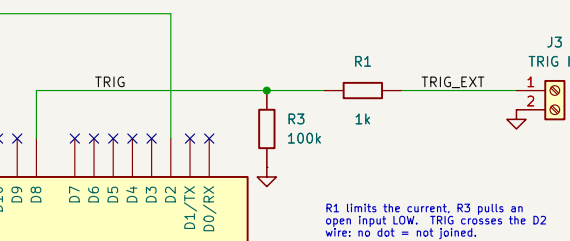
2 2.5 V REFERENCE

U4 makes a fixed 2.5 V (VREF) for the preamp and the ADC. C7 at its input, C5 at its output.

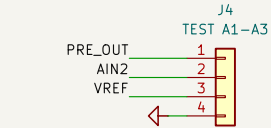
5 ADC (I2C 0x48)



4 TRIGGER INPUT



9 U3 POWER

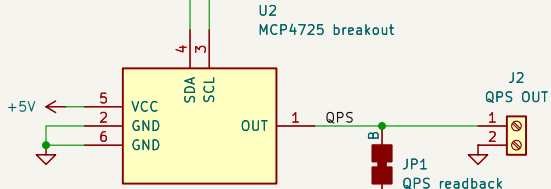


Scope/meter only: the names join these pins to the PRE_OUT, AIN2 and VREF wires.

8 ARDUINO UNO

USB: 5 V power and data to the laptop. D8 = trigger, D2 = ADC ready. A4/A5 carry the same SDA/SCL wires.

6 DAC -> QPS OUT (I2C 0x60)



0–5 V mass command to the quadrupole supply. JP1 (bridged) sends it back to ADC A2: cut to free A2.

OPERATING POINT (calculated DC values, +5V = 5.00 V; electron current: output rises)

Node / branch	I = 0	10 nA (min)	1 μA (max)	How
TIA_IN = J5 centre (U3 pin 8)	2.500 V	2.500 V	2.500 V	virtual ground = VREF (offset < 0.15 mV)
TIA_OUT (U3 pin 4)	2.500 V	2.520 V	4.500 V	VREF + I x 2.0 MΩ (R6)
PRE_OUT (ADC A1, J4–1)	2.500 V	2.520 V	4.499 V	TIA_OUT – R11 drop (< 1 mV)
A1 – A3 (what the ADC reads)	0 V	20 mV	2.000 V	I x 2.0 MΩ
ADC code, GAIN_TWO (±2.048 V)	0	320	32000	/ 62.5 μV per count; full scale 1.02 μA
ADC code, GAIN_ONE (±4.096 V)	0	160	16000	/ 125 μV per count
Current in R6 (= detector)	0	10 nA	1 μA	C2, C6: no DC current
U3 output current	–0.4 μA	–0.4 μA	–1.9 μA	R6 + ADC A1 input

Range limits: –1.09 μA at 4.75 V USB; 1.21 μA at 5.00 V (U3 output reaches V+ – 0.07 V).
ADC zero offset –±3 counts: measure with the beam off and subtract.

Fixed values	
+5V (UNO, USB)	5.00 V (4.75 – 5.25)
ADS_VDD (after Q1)	5.00 V, Q1 drop < 0.1 mV
VREF (U4 out)	2.500 V ±0.15 %
QPS / J2 (DAC out)	0 – 5.0 V, ≤ 0.5 mA into 10 kΩ
D8, 5 V trigger	4.95 V, 49.5 μA
D8, 3.3 V trigger	3.27 V, 32.7 μA (marginal)
SDA / SCL low	≤ 0.4 V, ~1.5 mA per line
RDY low	≤ 0.4 V, 0.46 mA
U3 LMP7721 supply	1.3 mA (1.7 max)
U1 ADS1115 supply	0.18 mA (0.3 max)
U2 MCP4725 supply	0.21 mA (0.4 max)
U4 REF3325	3.9 μA (5 max)
Shield total from 5 V	~1.7 mA typ, 6.2 mA worst

Firmware base: Grinias et al., J. Chem. Educ. 2016, 93, 1316
Order everything from Digi–Key: fab/digikey_bom.csv. All parts hand–soldered.
Team 5 scope: ADC side. MCP4725 mass command is fitted for Team 4's sequencer.
Portable quadrupole mass spectrometer – data acquisition and control block
MIE1001 Team 5 – Foundation of Imaging Engineering, Maastricht University (FSE, M4I)

Sheet: /	File: mie1001_daq_simple.kicad_sch	
Title: SIMPLE DAQ shield for Arduino UNO R3 – class breakout boards		
Size: A3	Date: 2026–10–01	Rev: J
KiCad E.D.A. 10.0.6		Id: 1/1